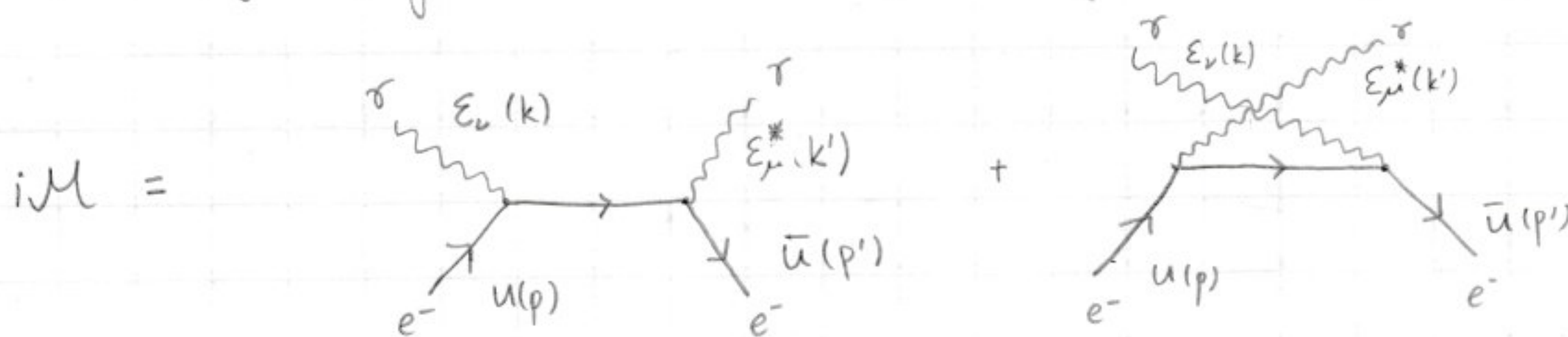


Exercise 7.7

Consider Compton scattering and derive the Klein-Nishina formula

$$\frac{d\sigma}{d\cos\theta} = \frac{\alpha^2 \pi}{m_e^2} \left[\frac{\omega'}{\omega} \right]^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta \right], \text{ where } \theta \text{ is the scattering angle in the lab frame}$$

We start by drawing the lowest order Feynman diagrams and using the Feynman rules:



By including the polarization and spin indices and dropping momentum indices since their information is in spin-index or polarization index, such that $u^r(p) = u^r$, $u^{r'}(p') = u^{r'}$, $\epsilon^\mu(k) = \epsilon^\mu$, $\epsilon^{\mu*}(k') = \epsilon^{\mu*}$

$$i\mathcal{M} = \bar{u}^{r'}(-ie\gamma^\mu)\epsilon_\mu^{s'*} \left[\frac{i(\not{p} + \not{k} + m_e)}{(p+k)^2 - m_e^2} \right] \epsilon_\nu^s(-ie\gamma^\nu)u^r + \bar{u}^{r'}(-ie\gamma^\nu)\epsilon_\nu^s \left[\frac{i(\not{p} - \not{k}' + m_e)}{(p-k')^2 - m_e^2} \right] \epsilon_\mu^{s'*}(-ie\gamma^\mu)u^r$$

we denote $m \equiv m_e$

$$p^2 = m^2, k^2 = 0$$

$$\Rightarrow (p+k)^2 - m^2 = 2(p \cdot k)$$

$$(p-k')^2 - m^2 = -2(p \cdot k')$$

$$\Rightarrow i\mathcal{M} = -ie^2 \epsilon_\mu^{s'*} \epsilon_\nu^s \bar{u}^{r'} \left[\frac{\gamma^\mu(\not{p} + \not{k} + m)\gamma^\nu}{2(p \cdot k)} + \frac{\gamma^\nu(\not{p} - \not{k}' + m)\gamma^\mu}{-2(p \cdot k')} \right] u^r$$

We can simplify further using Dirac algebra:

$$(\not{p} + m)\gamma^\nu u^r = [2p^\nu - \gamma^\nu \not{p} + \gamma^\nu m] u^r$$

$$= 2p^\nu u^r - \underbrace{\gamma^\nu \not{p}}_{=0} u^r$$

$$= 2p^\nu u^r$$

$$\not{p}\gamma^\nu = 2g^{\mu\nu}p_\mu - \gamma^\nu \not{p}p_\mu$$

$$\not{p}\gamma^\nu = 2p^\nu - \gamma^\nu \not{p}$$

$$\text{Dirac equation: } (\not{p} - m)u = 0$$

$$\Rightarrow i\mathcal{M} = -ie^2 \epsilon_\mu^{s'*} \epsilon_\nu^s \bar{u}^{r'} \left(\frac{\gamma^\mu \not{k} \gamma^\nu + 2\gamma^\mu p^\nu}{2(p \cdot k)} + \frac{\gamma^\nu \not{k}' \gamma^\mu - 2\gamma^\nu p^\mu}{2(p \cdot k')} \right) u^r$$

Now we look for simplification for adjoint of $\mathcal{M}$:

$$\mathcal{M}^\dagger = -e^2 \epsilon_\nu^{s'} \epsilon_\mu^{s'*} \bar{u}^r \left[\frac{\gamma^\nu(\not{p} + \not{k} + m)\gamma^\mu}{2(p \cdot k)} + \frac{\gamma^\mu(\not{p} - \not{k}' + m)\gamma^\nu}{-2(p \cdot k')} \right] u^{r'}$$

γ^ν & γ^μ swap compared to $\mathcal{M}$ because polarization is conjugated too: $\epsilon_\nu^s \rightarrow \epsilon_\nu^{s'}$

We simplify this using adjoint Dirac equation: $\bar{u}(\not{p} + m) = 0$ where $\not{p}$ denotes that it acts to the left and is $\not{p} = +p_\mu^\dagger \gamma^\mu = -p_\mu \gamma^\mu = -\not{p}$

$$\begin{aligned}\bar{u}(p)\gamma^\sigma[\not{p}+m] &= \bar{u}^\sigma [2p^\sigma - \not{p}\gamma^\sigma + m\gamma^\sigma] \quad \parallel -\not{p} = p_\mu^\dagger \gamma^\mu = \overleftarrow{\not{p}} \\ &= \bar{u}^\sigma 2p^\sigma + \underbrace{\bar{u}^\sigma [\overleftarrow{\not{p}} + m]}_{=0} \gamma^\sigma \\ &= \bar{u}^\sigma 2p^\sigma, \quad \text{which means}\end{aligned}$$

$$M^\dagger = -e^2 \varepsilon_\sigma^{\dagger} \varepsilon_\rho^{\dagger} \bar{u}^\sigma \left[\frac{\gamma^\sigma \not{k} \gamma^\rho + 2\gamma^\sigma p^\rho}{2(p \cdot k)} + \frac{\gamma^\rho \not{k}' \gamma^\sigma - 2\gamma^\sigma p^\rho}{2(p \cdot k')} \right] u^{\rho'}$$

To get the full expression we must also average over spin and polarization:

$$|M_{fi}|^2 = \langle |M|^2 \rangle = \frac{1}{4} \sum_{ss'rr'} |M|^2 = \frac{1}{4} \sum_{ss'rr'} M M^\dagger$$

$$|M_{fi}|^2 = \frac{e^4}{4} \sum_{ss'rr'} \varepsilon_\mu^{\dagger} \varepsilon_\nu^{\dagger} \varepsilon_\sigma^{\dagger} \varepsilon_\rho^{\dagger} (\bar{u}^{r'} [\not{p}_{prop}]^{\mu\nu} u^r) (\bar{u}^r [\not{p}_{prop}^\dagger]^{\rho\sigma} u^{r'})$$

$$\text{where } [\not{p}_{prop}]^{\mu\nu} = \left[\frac{\gamma^\mu \not{k} \gamma^\nu + 2\gamma^\mu p^\nu}{2(p \cdot k)} + \frac{\gamma^\nu \not{k}' \gamma^\mu - 2\gamma^\nu p^\mu}{2(p \cdot k')} \right]$$

$$\text{and } [\not{p}_{prop}^\dagger]^{\rho\sigma} = \left[\frac{\gamma^\sigma \not{k} \gamma^\rho + 2\gamma^\sigma p^\rho}{2(p \cdot k)} + \frac{\gamma^\rho \not{k}' \gamma^\sigma - 2\gamma^\sigma p^\rho}{2(p \cdot k')} \right]$$

Doing the sum over polarization and writing $|M_{fi}|^2$ in component form (with repeat ^{latin} index = _{= conventional sum}):

$$|M_{fi}|^2 = \frac{e^4}{4} \sum_{rr'} g_{\mu\rho} g_{\nu\sigma} \left(\bar{u}^{r'}_a [\not{p}_{prop}]^{\mu\nu}_{ab} u^r_b \bar{u}^r_c [\not{p}_{prop}^\dagger]^{\rho\sigma}_{cd} u^{r'}_d \right) \quad \parallel \text{ we use } \sum u^{r'}_a \bar{u}^{r'}_a = \not{p}' + m$$

$$= \frac{e^4}{4} g_{\mu\rho} g_{\nu\sigma} \underbrace{\left([\not{p}_{prop}]^{\mu\nu}_{ab} \sum_r u^r_b \bar{u}^r_c [\not{p}_{prop}^\dagger]^{\rho\sigma}_{cd} \sum_{r'} u^{r'}_d \bar{u}^{r'}_a \right)}_{= \text{Trace}}$$

$$= \frac{e^4}{4} g_{\mu\rho} g_{\nu\sigma} \text{Tr} \left[[\not{p}_{prop}]^{\mu\nu} (\not{p} + m) [\not{p}_{prop}^\dagger]^{\rho\sigma} (\not{p}' + m) \right]$$

$$|M_{fi}|^2 = \frac{e^4}{4} \text{Tr} \left[(\not{p}' + m) \left[\frac{\gamma^\nu \not{k} \gamma^\mu + 2\gamma^\nu p^\mu}{2(p \cdot k)} + \frac{\gamma^\mu \not{k}' \gamma^\nu - 2\gamma^\mu p^\nu}{2(p \cdot k')} \right] (\not{p} + m) \left[\frac{\gamma_\nu \not{k} \gamma_\mu + 2\gamma_\nu p_\mu}{2(p \cdot k)} + \frac{\gamma_\mu \not{k}' \gamma_\nu - 2\gamma_\mu p_\nu}{2(p \cdot k')} \right] \right]$$

$$= \frac{e^4}{4} \left[\frac{T_1}{(2p \cdot k)^2} + \frac{T_2}{(2p \cdot k')^2} + \frac{T_3}{(2p \cdot k)(2p \cdot k')} + \frac{T_4}{(2p \cdot k')(2p \cdot k)} \right], \quad \text{where}$$

$$T_1 = \text{Tr} [(\not{p}' + m) (\gamma^\mu \not{k} \gamma^\nu + 2 \gamma^\mu p^\nu) (\not{p} + m) (\gamma_\nu \not{k} \gamma_\mu + 2 \gamma_\mu p_\nu)]$$

$$T_2 = \text{Tr} [(\not{p}' + m) (\gamma^\nu \not{k} \gamma^\mu - 2 \gamma^\nu p^\mu) (\not{p} + m) (\gamma_\mu \not{k} \gamma_\nu - 2 \gamma_\nu p_\mu)] = T_1 (k \rightarrow -k')$$

$$T_3 = \text{Tr} [(\not{p}' + m) (\gamma^\mu \not{k} \gamma^\nu + 2 \gamma^\mu p^\nu) (\not{p} + m) (\gamma_\mu \not{k} \gamma_\nu - 2 \gamma_\nu p_\mu)]$$

$$T_4 = \text{Tr} [(\not{p}' + m) (\gamma^\nu \not{k} \gamma^\mu - 2 \gamma^\nu p^\mu) (\not{p} + m) (\gamma_\nu \not{k} \gamma_\mu + 2 \gamma_\mu p_\nu)] = T_3 (k)$$

Same as T_1 but with k changed to $-k'$

$\mu \leftrightarrow \nu$ swap/renameing doesn't matter since they are contracted fully inside the trace

for T_4 we don't need to change T_3 from $+k$ to $-k'$ ^{in the argument} because the multiplication of arguments stays same under the change $k \rightarrow -k'$ & $k' \rightarrow -k$.

$\Rightarrow$ We only need to solve trace T_1 and T_4 :

$$T_1 = \text{Tr} [(\not{p}' + m) (\gamma^\mu \not{k} \gamma^\nu + 2 \gamma^\mu p^\nu) (\not{p} + m) (\gamma_\nu \not{k} \gamma_\mu + 2 \gamma_\mu p_\nu)]$$

of the 16 component half of them give trace of 0 since they have odd number of γ -matrices which means the non-zero parts are:

$$T_1 = \text{Tr} [\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\nu \not{k} \gamma_\mu + 2 \not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\mu p_\nu + 2 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\nu \not{k} \gamma_\mu + 4 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\mu p_\nu + m^2 \gamma^\mu \not{k} \gamma^\nu \gamma_\nu \not{k} \gamma_\mu + 2m^2 \gamma^\mu \not{k} \gamma^\nu \gamma_\mu p_\nu + 2m^2 \gamma^\mu p^\nu \gamma_\nu \not{k} \gamma_\mu + 4m^2 \gamma^\mu p^\nu \gamma_\mu p_\nu]$$

Let's calculate these traces individually using trace technology from chapter 5.1 in book (Peskin & Schroeder)

$$\begin{aligned} \text{Tr} [\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\nu \not{k} \gamma_\mu] &= -2 \text{Tr} [\not{p}' \gamma^\mu \not{k} \not{p} \not{k} \gamma_\mu] = 4 \text{Tr} [\not{p}' \not{k} \not{p} \not{k}] \\ &= -2 \not{p}' \not{k} \not{p} \not{k} \\ &= 4 [(p' \cdot k)(p \cdot k) - (p' \cdot p)(k^2) + (p' \cdot k)(k \cdot p)] \\ \Rightarrow \text{Tr} [\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\nu \not{k} \gamma_\mu] &= \underline{16 \cdot 2 (p' \cdot k)(p \cdot k)} \end{aligned}$$

$$\text{Tr} [2 \not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\mu p_\nu] = -4 \text{Tr} [\not{p}' \not{p} \not{k}] = -4m^2 \text{Tr} [\not{p}' \not{k}] = \underline{-16m^2 (p' \cdot k)}$$

$$\text{Tr} [2 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\nu \not{k} \gamma_\mu] = 2 \text{Tr} [\not{p}' \gamma^\mu \not{p} \not{k} \gamma_\mu] = -4m^2 \text{Tr} [\not{p}' \not{k}] = \underline{-16m^2 (p' \cdot k)}$$

$$\text{Tr} [4 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\mu p_\nu] = 4m^2 \text{Tr} [\not{p}' \gamma^\mu \not{p} \gamma_\mu] = -8m^2 \text{Tr} [\not{p}' \not{p}] = \underline{-32m^2 (p' \cdot p)}$$

$$m^2 \text{Tr} [\gamma^\mu \not{k} \gamma^\nu \gamma_\nu \not{k} \gamma_\mu] = 4m^2 \text{Tr} [\gamma^\mu \not{k} \not{k} \gamma_\mu] = \underline{0} \quad // \quad k \cdot k = k^2 = 0$$

$$2m^2 \text{Tr} [\gamma^\mu \not{k} \gamma^\nu \gamma_\mu p_\nu] = 2m^2 \text{Tr} [\gamma_\mu \gamma^\mu \not{k} \not{p}] = 8m^2 \text{Tr} [\not{k} \not{p}] = \underline{32m^2 (p \cdot k)}$$

$$2m^2 \text{Tr} [\gamma^\mu p^\nu \gamma_\nu \not{k} \gamma_\mu] = 2m^2 \text{Tr} [\gamma_\mu \gamma^\mu \not{p} \not{k}] = \underline{32m^2 (p \cdot k)}$$

$$4m^2 \text{Tr} [\gamma^\mu p^\nu \gamma_\mu p_\nu] = 4m^2 \text{Tr} [\gamma^\mu \gamma_\mu p^2] = 16m^2 \text{Tr} (m^2) = 16 \cdot 4m^4$$

$$\Rightarrow T_1 = 16 [4m^4 - 2m^2 p \cdot p' + 4m^2 p \cdot k - 2m^2 p' \cdot k + 2(p \cdot k)(p' \cdot k)]$$

$$T_3 = \text{Tr} \left[(\not{p}' + m) (\gamma^\mu \not{k} \gamma^\nu + 2\gamma^\mu p^\nu) (\not{p} + m) (\gamma_\mu \not{k} \gamma_\nu - 2\gamma_\nu p_\mu) \right]$$

The non-zero parts (even number of γ -matrices) are

$$T_3 = \text{Tr} \left[\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\mu \not{k} \gamma_\nu - 2 \not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\nu p_\mu + 2 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\mu \not{k} \gamma_\nu - 4 \not{p}' \gamma^\mu p^\nu \not{p} \gamma_\nu p_\mu \right. \\ \left. + m^2 \gamma^\mu \not{k} \gamma^\nu \gamma_\mu \not{k} \gamma_\nu - 2m^2 \gamma^\mu \not{k} \gamma^\nu \gamma_\nu p_\mu + 2m^2 \gamma^\mu p^\nu \gamma_\mu \not{k} \gamma_\nu - 4m^2 \gamma^\mu p^\nu \gamma_\nu p_\mu \right]$$

We calculate these individually:

$$\text{Tr} \left[\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\mu \not{k} \gamma_\nu \right] = -2 \text{Tr} \left[\not{p}' \not{p} \gamma^\nu \not{k} \gamma_\nu \right] = 0 \quad \left\| \not{k} \not{k} = k^2 = 0 \right.$$

$$-2 \text{Tr} \left[\not{p}' \gamma^\mu \not{k} \gamma^\nu \not{p} \gamma_\nu p_\mu \right] = +4 \text{Tr} \left[\not{p}' \not{p} \not{k} \not{p} \right] = 4 \cdot 4 \left[(p' \cdot p)(k \cdot p) - (p' \cdot k)p^2 + (p' \cdot p)(p \cdot k) \right] \\ = -16m^2 (p' \cdot k) + 16 [2 \cdot (p \cdot p)(p \cdot k)]$$

$$+2 \text{Tr} \left[\not{p}' \gamma^\mu p^\nu \not{p} \gamma_\mu \not{k} \gamma_\nu \right] = -4 \text{Tr} \left[\not{p}' \not{p} \not{k} \not{p} \right] = +16m^2 (p' \cdot k) - 16 [2 \cdot (p' \cdot p)(p \cdot k)]$$

$$-4 \text{Tr} \left[\not{p}' \gamma^\mu p^\nu \not{p} \gamma_\nu p_\mu \right] = -4 \text{Tr} \left[\not{p}' \not{p} \not{p} \not{p} \right] = -4m^2 \cdot 4 (p' \cdot p) = -16m^2 (p' \cdot p)$$

$$+m^2 \text{Tr} \left[\gamma^\mu \not{k} \gamma^\nu \gamma_\mu \not{k} \gamma_\nu \right] = -4m^2 \text{Tr} \left[\not{k} \not{k} \right] = 0$$

$$-2m^2 \text{Tr} \left[\gamma^\mu \not{k} \gamma^\nu \gamma_\nu p_\mu \right] = -8m^2 \text{Tr} \left[\not{p} \not{k} \right] = -32m^2 (p \cdot k)$$

$$+2m^2 \text{Tr} \left[\gamma^\mu p^\nu \gamma_\mu \not{k} \gamma_\nu \right] = 8m^2 \text{Tr} \left[\not{k} \not{p} \right] = 32m^2 (p \cdot k)$$

$$-4m^2 \text{Tr} \left[\gamma^\mu p^\nu \gamma_\nu p_\mu \right] = -4m^2 \text{Tr} \left[\not{p} \not{p} \right] = -16m^4$$

$$\Rightarrow T_3 = 16 [-m^4 - m^2(p' \cdot p)] = -16 [m^4 + m^2(p' \cdot p)] = -8 [2m^4 + 2m^2(p' \cdot p)]$$

Let's now write the traces T_1 and T_3 in terms of Mandelstam variables:

$$s = (p+k)^2 = 2p \cdot k + m^2 = 2p' \cdot k' + m^2, \quad t = (p'-p)^2 = -2p \cdot p' + 2m^2 = -2k \cdot k'$$

$$u = (k'-p)^2 = -2k' \cdot p + m^2 = -2k \cdot p' + m^2 \quad \text{and we have } s+t+u = 2m^2$$

$$\Rightarrow T_3 = -8 [2m^4 + m^2(-t + 2m^2)] = -8 [4m^4 + (s+u-m^2-m^2)m^2]$$

$$T_3 = -8 [4m^4 + m^2(s-m^2) + m^2(u-m^2)] = T_4 \quad \text{Similarly for } T_1 \text{ we get}$$

$$T_1 = 16 [2m^4 + m^2(s-m^2) - \frac{1}{2}(s-m^2)(u-m^2)] \quad \left\| \text{doing } k \rightarrow -k' \text{ makes } s \rightarrow u \text{ \& } u \rightarrow s \right. \\ \left. \text{which gives us } T_2 \right.$$

$$\Rightarrow T_2 = 16 [2m^4 + m^2(u-m^2) - \frac{1}{2}(s-m^2)(u-m^2)]$$

The matrix-element squared term is then (in terms of Mandelstam-variables)

$$|M_{fi}|^2 = \frac{e^4}{4} \left[\frac{T_1}{(s-m^2)^2} + \frac{T_2}{(u-m^2)^2} - \frac{T_3}{(s-m^2)(u-m^2)} - \frac{T_4}{(s-m^2)(u-m^2)} \right] \quad // \quad T_3 = T_4$$

$$= \frac{e^4}{4} \left[\frac{T_1}{(s-m^2)^2} + \frac{T_2}{(u-m^2)^2} - \frac{2T_3}{(s-m^2)(u-m^2)} \right]$$

$$\frac{T_1}{(s-m^2)^2} = \frac{16 \left[2m^4 + m^2(s-m^2) - \frac{1}{2}(s-m^2)(u-m^2) \right]}{(s-m^2)^2}$$

$$= \frac{32m^4}{(s-m^2)^2} + 16 \left[\frac{m^2 - \frac{1}{2}(u-m^2)}{(s-m^2)} \right]$$

$$\frac{T_2}{(u-m^2)^2} = \frac{16 \left[2m^4 + m^2(u-m^2) - \frac{1}{2}(s-m^2)(u-m^2) \right]}{(u-m^2)^2}$$

$$= \frac{32m^4}{(u-m^2)^2} + 16 \left[\frac{m^2 - \frac{1}{2}(s-m^2)}{(u-m^2)} \right]$$

$$- \frac{2T_3}{(s-m^2)(u-m^2)} = \frac{16 \left[4m^4 + m^2(s-m^2) + m^2(u-m^2) \right]}{(s-m^2)(u-m^2)}$$

$$= 2 \cdot \frac{32m^4}{(s-m^2)(u-m^2)} + 16 \left[\frac{m^2}{(u-m^2)} \right] + 16m^2 \left[\frac{1}{(s-m^2)} \right]$$

Summing these we get:

$$|M_{fi}|^2 = e^4 \left[8m^4 \left(\frac{1}{(s-m^2)^2} + \frac{1}{(u-m^2)^2} + \frac{2}{(s-m^2)(u-m^2)} \right) + 4m^2 \left(\frac{2}{(u-m^2)} + \frac{2}{(s-m^2)} \right) - \frac{2(u-m^2)}{(s-m^2)} - \frac{2(s-m^2)}{(u-m^2)} \right]$$

$$|M_{fi}|^2 = 2e^4 \left[4m^4 \left[\frac{1}{(s-m^2)} + \frac{1}{(u-m^2)} \right]^2 + 4m^2 \left[\frac{1}{(s-m^2)} + \frac{1}{(u-m^2)} \right] - \left[\frac{u-m^2}{s-m^2} + \frac{s-m^2}{u-m^2} \right] \right]$$

Now we plug back in $(s-m^2) = 2p \cdot k$ and $(u-m^2) = -2p \cdot k'$

$$|M_{fi}|^2 = 2e^4 \left[4m^4 \left(\frac{1}{2p \cdot k} - \frac{1}{2p \cdot k'} \right)^2 + 4m^2 \left[\frac{1}{2p \cdot k} - \frac{1}{2p \cdot k'} \right] - \left[\frac{-p \cdot k'}{p \cdot k} - \frac{p \cdot k}{p \cdot k'} \right] \right]$$

$$\Rightarrow |M_{fi}|^2 = 2e^4 \left[\frac{p \cdot k'}{p \cdot k} + \frac{p \cdot k}{p \cdot k'} + 2m^2 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right) + m^4 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right)^2 \right]$$

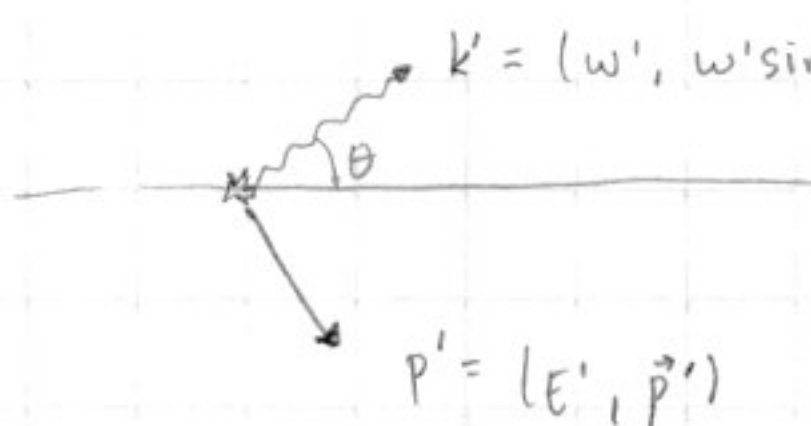
Now this is our desired spin-averaged matrix-element squared and we use this to calculate the cross section.

In the lab-frame we have the following situation:

Before:

$$k = (w, w\hat{z}) \quad p = (m, 0)$$

After:

$$k' = (w', w' \sin \theta, 0, w' \cos \theta)$$


$$p' = (E', \vec{p}')$$

Since we express cross section in terms of w, θ we need to find out w' :

$$m^2 = (p')^2 = (p + k - k')^2 = p^2 + 2p \cdot (k - k') - 2k \cdot k'$$

$$= m^2 + 2m(w - w') - 2ww'(1 - \cos \theta)$$

$$\Rightarrow \frac{1}{w'} - \frac{1}{w} = \frac{1}{m}(1 - \cos \theta)$$

$$\Rightarrow w' = \frac{w}{1 + \frac{w}{m}(1 - \cos \theta)}$$

The two-particle phase space is

$$d\pi_2 = \left(\prod_{i=1}^2 \frac{d^3 p_i}{(2\pi)^3 2E_i} \right) (2\pi)^4 \delta^{(4)}(k + p - p' - k')$$

The phase space integral is

$$\int d\pi_2 = \int \frac{d^3 k'}{(2\pi)^3} \frac{1}{2w'} \frac{d^3 p'}{(2\pi)^3} \frac{1}{2E'} (2\pi)^4 \delta(w + m - w' - E') \delta^{(3)}(\vec{k} - \vec{p}' - \vec{k}')$$

We integrate over p' : δ -function gives $\vec{p}' = \vec{k} - \vec{k}'$

$$= \int \frac{d^3 k'}{(2\pi)^3} 2\pi \frac{1}{2w' 2E'} \delta(w + m - w' - \sqrt{m^2 + w^2 + (w')^2 - 2ww' \cos \theta})$$

$\Rightarrow E' = \sqrt{m^2 + w^2 + (w')^2 - 2ww' \cos \theta}$
 // Move to spherical coordinates
 $d^3 k' = (w')^2 dw' d\Omega$

$$= \int \frac{dw' d\Omega}{(2\pi)^2} \frac{w'}{4E'} \delta(w + m - w' - \sqrt{m^2 + w^2 + w'^2 - 2ww' \cos \theta})$$

Using $\delta(g(x)) = \frac{\delta(x - x_0)}{|g'(x_0)|}$ and integrating over $d\phi$ we get

$$\int d\pi_2 = \int \frac{d\cos \theta}{2\pi} \frac{w'}{4E'} \frac{1}{\left| 1 + \frac{w' - w \cos \theta}{E'} \right|}$$

In this case $x_0 = w'_0 = -E' + w + m$

$\Rightarrow \delta$ -function gives $E' = w + m - w'$

$$= \int \frac{d\cos \theta}{8\pi} \frac{w'}{|E' + w' - w \cos \theta|} = \frac{1}{8\pi} \int d\cos \theta \frac{w'}{|w + m - w' + w' - w \cos \theta|}$$

$$= \frac{1}{8\pi} \int d\cos \theta \frac{w'}{|m + w(1 - \cos \theta)|} = \frac{1}{8\pi} \int d\cos \theta \frac{w'}{m + w(1 - \cos \theta)}$$

using

$$\omega' = \frac{\omega}{1 + \frac{\omega}{m}(1 - \cos\theta)}$$

, we get.

$$\int d\pi_2 = \frac{1}{8\pi} \int d\cos\theta \frac{(\omega')^2}{\omega m}$$

Now we plug this into general cross-section formula (4.79 in Peskin & Schroeder) and setting $|v_A - v_B| = 1$, we get

$$\frac{d\sigma}{d\cos\theta} = \frac{1}{2\omega} \frac{1}{2m} \cdot \frac{1}{8\pi} \frac{(\omega')^2}{\omega m} \cdot |M_{fi}|^2$$

We have expression for $|M_{fi}|^2$, and we can get it in terms of ω, ω' by writing $p \cdot k = m\omega$ and $p \cdot k' = m\omega'$

$$\Rightarrow |M_{fi}|^2 = 2e^4 \left[\frac{m\omega'}{m\omega} + \frac{m\omega}{m\omega'} + 2m^2 \left(\frac{1}{m\omega} - \frac{1}{m\omega'} \right) + m^2 \left(\frac{1}{m\omega} - \frac{1}{m\omega'} \right)^2 \right]$$

$$= 2e^4 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 2m \left(\frac{1}{\omega} - \frac{1}{\omega'} \right) + m^2 \left(\frac{1}{\omega} - \frac{1}{\omega'} \right)^2 \right]$$

$$\Rightarrow \frac{d\sigma}{d\cos\theta} = \frac{2e^4}{2\omega \cdot 2m \cdot 8\pi} \frac{(\omega')^2}{\omega m} \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 2m \left(\frac{1}{\omega} - \frac{1}{\omega'} \right) + m^2 \left(\frac{1}{\omega} - \frac{1}{\omega'} \right)^2 \right]$$

We note from before that $\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1}{m}(1 - \cos\theta)$

$$\frac{d\sigma}{d\cos\theta} = \frac{e^4}{16\pi^2 m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2m \frac{1}{m}(1 - \cos\theta) + m^2 \left[\frac{1}{m}(1 - \cos\theta) \right]^2 \right]$$

$$= \frac{e^4}{16\pi^2 m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2(1 - \cos\theta) + (1 - 2\cos\theta + \cos^2\theta) \right] \quad \parallel e^2 = 4\pi\alpha$$

$$= \frac{\alpha^2 \pi}{m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 1 + \cos^2\theta \right] \quad \parallel \cos^2\theta - 1 = -\sin^2\theta$$

$$\frac{d\sigma}{d\cos\theta} = \frac{\alpha^2 \pi}{m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta \right] \quad \square$$

We have derived the Klein-Nishina formula in the lab frame.

In the limit $\omega \rightarrow 0$ we get

$$\lim_{\omega \rightarrow 0} \frac{\omega'}{\omega} = \lim_{\omega \rightarrow 0} \frac{1}{1 + \frac{\omega}{m}(1 - \cos\theta)} = 1 \quad \text{and} \quad \lim_{\omega \rightarrow 0} \frac{\omega}{\omega'} = \lim_{\omega \rightarrow 0} \frac{\left[1 + \frac{\omega}{m}(1 - \cos\theta) \right] \omega}{\omega} = 1$$

$$\Rightarrow \lim_{\omega \rightarrow 0} \frac{d\sigma}{d\cos\theta} = \frac{\alpha^2 \pi}{m^2} \left[2 - \sin^2\theta \right] = \frac{\alpha^2 \pi}{m^2} \left[1 + \cos^2\theta \right]$$

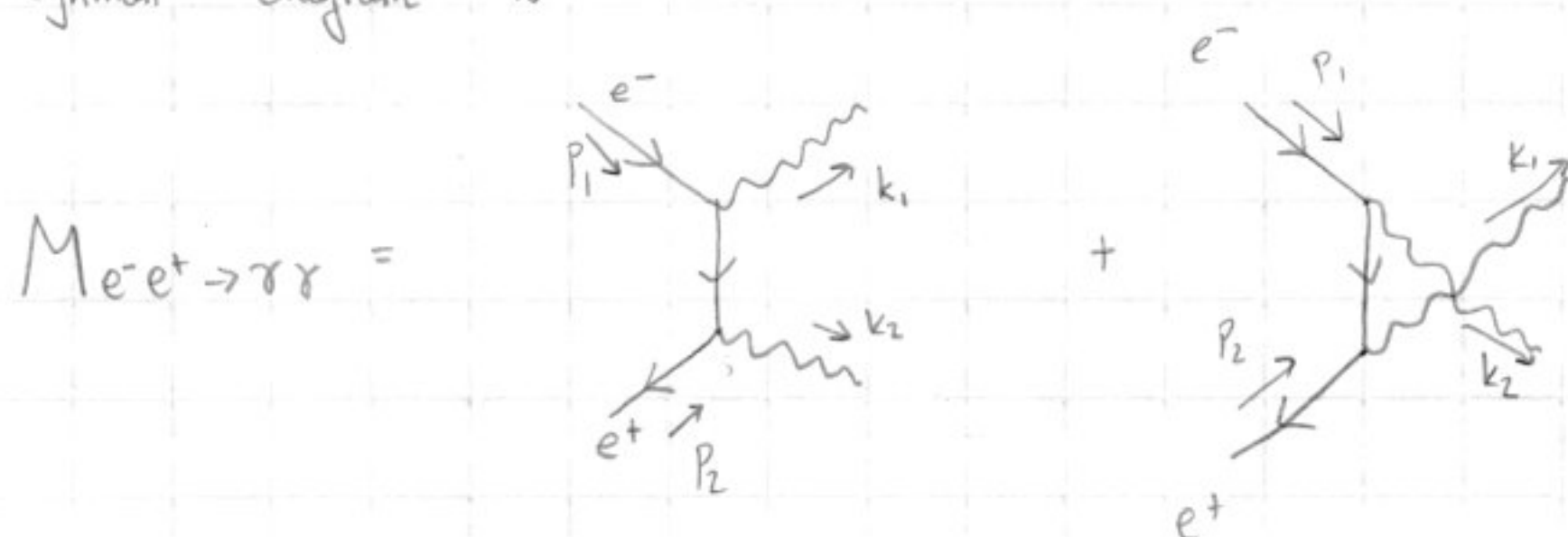
The differential cross section in the limit $\omega \rightarrow 0$ is

$$\frac{d\sigma}{d\cos\theta} = \frac{\alpha^2 \pi}{m^2} [1 + \cos^2\theta]$$

Exercise 7.8

Look at reaction $e^+e^- \rightarrow \gamma\gamma$. Use crossing symmetry to get the spin-averaged matrix-element squared.

The Feynman diagram is



By crossing symmetry this is related to Compton scattering. We get amplitudes for this via substitution to $M_{e\gamma \rightarrow e\gamma}$ matrix:

$$p \rightarrow p_1, \quad p' \rightarrow -p_2, \quad k \rightarrow -k_1, \quad k' \rightarrow k_2$$

Making these substitution to spin & polarization averaged Matrix Element squared of $M_{e\gamma \rightarrow e\gamma}$ we get

$$\langle |M_{e^+e^- \rightarrow \gamma\gamma}|^2 \rangle = |M_{fi}|^2 = -2e^4 \left[\frac{p_1 \cdot k_2}{p_1 \cdot k_1} + \frac{p_1 \cdot k_1}{p_1 \cdot k_2} + 2m^2 \left(\frac{1}{p_1 \cdot k_1} + \frac{1}{p_1 \cdot k_2} \right) - m^4 \left(\frac{1}{p_1 \cdot k_1} + \frac{1}{p_1 \cdot k_2} \right)^2 \right]$$

The overall sign comes from crossing relation

$$\sum u(p) \bar{u}(p) = \not{p} + m = -(\not{k} - m) = -\sum v(k) \bar{v}(k)$$

So we remove it. Since we cancel by hand one minus sign for each crossed fermion:

$$\Rightarrow \frac{1}{4} \sum |M_{e^+e^- \rightarrow \gamma\gamma}|^2 = 2e^4 \left[\frac{p_1 \cdot k_2}{p_1 \cdot k_1} + \frac{p_1 \cdot k_1}{p_1 \cdot k_2} + 2m^2 \left[\frac{1}{p_1 \cdot k_1} + \frac{1}{p_1 \cdot k_2} \right] - m^4 \left[\frac{1}{p_1 \cdot k_1} + \frac{1}{p_1 \cdot k_2} \right]^2 \right] \quad \square$$

This is the spin averaged matrix element for $e^+e^- \rightarrow \gamma\gamma$ process.